

Comprehensive Technical Report:

Neural-GPS

Rat Mind Reading and Decoding Hippocampal Place Cells

Developer: [Ali](#)

GitHub Repository: [Neural-GPS](#)

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Abstract

This report provides an in-depth analysis of the Neural-GPS project, designed to simulate the activity of place cells in the rat hippocampus and decode its spatial position. In this document, we examine the complete cycle of converting physical stimuli into neural codes (Encoding) and the reverse process (Decoding) using advanced statistical models, Shannon's information theory, dimensionality reduction (PCA), and Bayesian inference. The objective of this report is to present a precise and comprehensible mathematical framework for understanding spatial representation mechanisms in the mammalian brain.

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1 Introduction and Problem Statement

The hippocampus is a key structure in the mammalian brain that plays a fundamental role in memory and spatial navigation. The discovery of "place cells" by John O'Keefe (which led to the 2014 Nobel Prize) demonstrated that the brain possesses an internal positioning system (GPS).

In the **Neural-GPS** project, we assume a rat is navigating along a 1D linear track with a length of 100 cm ($x \in [0, 100]$). Simultaneously, the spike train activity of 50 neurons from its hippocampus is recorded. The main challenge in this project is the mathematical formulation of this system across two overarching phases:

- **Encoding:** How is the continuous variable of position transformed into discrete spike trains?
- **Decoding:** How can we estimate the exact position of the rat at any given moment by observing the activity pattern of these 50 neurons?

2 Phase 1: Neural Encoding Modeling with GLM

In computational neuroscience, neuronal activity is a stochastic yet structured phenomenon. To simulate this behavior, Generalized Linear Models (GLMs) are employed, allowing us to combine Poisson noise with nonlinear dependencies.

2.1 Firing Rate Equation

Let $\lambda_i(t)$ be the expected firing rate of neuron i at time t . We model this rate using an exponential link function as follows:

$$\lambda_i(t) = \exp \left(c - \frac{(x(t) - \mu_i)^2}{2\sigma^2} + h \cdot y_i(t-1) \right)$$

This equation consists of three essential components:

1. **Receptive Field:** The term $\frac{(x(t) - \mu_i)^2}{2\sigma^2}$ represents a Gaussian bell curve. μ_i is the preferred location of the neuron (center of the receptive field), and σ is the width of this field. The closer the rat is to μ_i , the smaller this term becomes, resulting in a larger output from the exponential function.

2. **History Effect and Refractory Period:** The variable $y_i(t - 1)$ indicates the occurrence of a spike in the previous time step. The coefficient h (usually a large negative number) ensures that if the neuron has recently fired, its probability of firing again is significantly reduced.
3. **Spike Generation:** The number of spikes in a time interval Δt follows a Poisson distribution:

$$P(s_i|\lambda_i) = \frac{(\lambda_i \Delta t)^{s_i} e^{-\lambda_i \Delta t}}{s_i!}$$

3 Phase 2: Mutual Information Analysis

Not all neurons in the network are equally useful for position estimation. To quantify the informational value of each neuron, we utilize Shannon’s information theory.

3.1 Entropy and Uncertainty Reduction

The mutual information between the rat’s position (X) and the neuron’s firing rate (R_i) is defined as:

$$I(X; R_i) = \sum_{x \in X} \sum_{r \in R_i} P(x, r) \log_2 \left(\frac{P(x, r)}{P(x)P(r)} \right)$$

This equation calculates how much our uncertainty (entropy) regarding the rat’s position is reduced by knowing the firing status of neuron i . Neurons with a very sharp receptive field (small σ) generally possess higher mutual information, as their firing signifies the rat’s definitive presence at a specific location.

4 Phase 3: Discovering the Neural Manifold with PCA

The activity space of 50 neurons is a 50-dimensional space. However, since the external stimulus (position) is only one-dimensional, the network’s dynamics are expected to lie on a lower-dimensional manifold within this 50-dimensional space.

4.1 Principal Component Analysis (PCA)

Using the covariance matrix of the smoothed signals, eigenvectors and eigenvalues are computed:

$$\Sigma \mathbf{v} = \lambda \mathbf{v}$$

By projecting the data onto the first two principal components (PC1 and PC2), the underlying geometric structure of the neural activity is revealed, which typically exhibits a continuous topology corresponding to the physical track.

5 Phase 4: Bayesian Decoding

In this stage, the goal is to find the most probable position of the rat given the neural activity vector $S = [s_1, \dots, s_{50}]$ within a specific time window.

5.1 Bayes' Rule in Neural Decoding

Using Bayes' rule, we have:

$$P(X = x|S) = \frac{P(S|X = x) \cdot P(X = x)}{P(S)}$$

- **Likelihood:** Assuming independent neuronal firing, the probability of observing vector S given the rat is at position x is the product of Poisson distributions:

$$P(S|X = x) = \prod_{i=1}^{50} \frac{(\lambda_i(x)\Delta t)^{s_i} e^{-\lambda_i(x)\Delta t}}{s_i!}$$

- **Prior:** To prevent illogical predictions (such as the rat suddenly teleporting from one end of the track to the other), we utilize a normal distribution based on the previous position:

$$P(X_t = x) \sim \mathcal{N}(x_{t-1}, \sigma_{speed}^2)$$

Finally, using the **Maximum A Posteriori (MAP)** method, the point at which the posterior function is maximized is selected as the estimated position:

$$\hat{x}_{MAP} = \arg \max_x P(X = x|S)$$

6 Simulation Results Analysis (Figure Review)

In this section, we conduct a detailed analysis of the project’s visual outputs, as presented in Figure 1.

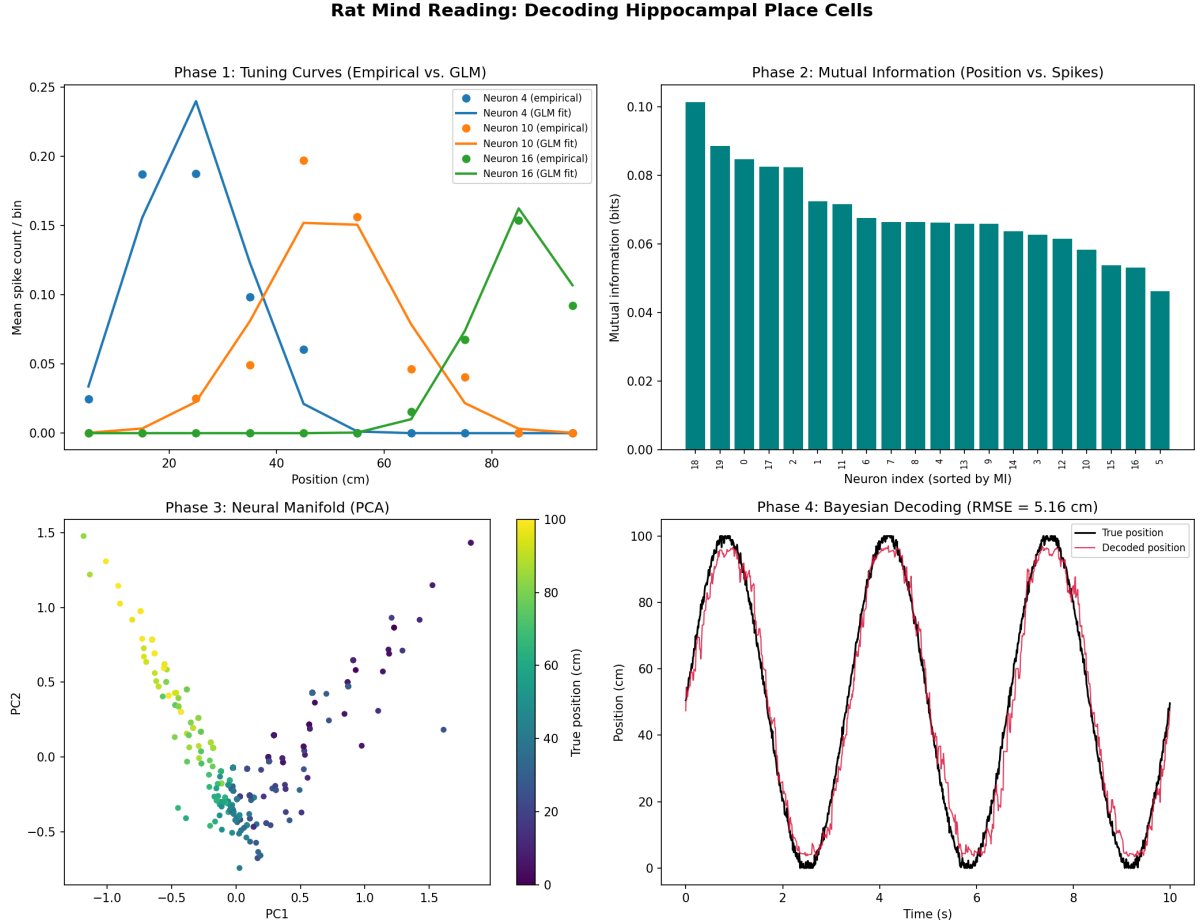


Figure 1: Comprehensive simulation results of rat mind reading across 4 phases. (Top Left: Tuning Curves, Top Right: Mutual Information, Bottom Left: PCA Neural Manifold, Bottom Right: Bayesian Decoding Result)

6.1 Analysis of the Four Subplots

1. **Phase 1 - Tuning Curves:** In the top-left chart, we observe that three distinct neurons (Neurons 4, 10, and 16) each exhibit peak activity at specific points along the track (approximately 25, 55, and 85 cm). The alignment of empirical data points with the fitted GLM lines demonstrates the high accuracy of the Gaussian-Poisson model in simulating receptive fields.

2. **Phase 2 - Mutual Information:** The bar chart in the top-right ranks the neurons based on the amount of information they provide about the rat’s location. Neuron 18, yielding over 0.10 bits of information, is the most valuable neuron in this time window. This variation in informational value highlights the importance of feature selection in BCI (Brain-Computer Interface) networks.
3. **Phase 3 - Neural Manifold (PCA):** In the bottom-left plot, the 50-dimensional data has been reduced to 2 dimensions. The points are color-coded based on the rat’s true position (from purple for 0 cm to yellow for 100 cm). The V-shaped structure formed in this space illustrates that the neural network has created a highly ordered and continuous topological representation of the 1D physical space.
4. **Phase 4 - Bayesian Decoding:** The bottom-right plot represents the pinnacle of the project’s success. The black curve shows the rat’s actual sinusoidal movement over time, while the red curve is our Bayesian model’s estimation based purely on reading the spikes. Tracking the trajectory so closely, with a Root Mean Square Error (RMSE) of 5.16 cm, proves that Bayesian inference successfully filtered out the Poisson noise and decoded the position with remarkable precision.

7 Conclusion

The **Neural-GPS** project successfully demonstrated how powerful tools from statistics and machine learning can be utilized to achieve a profound understanding of neural codes. The combination of GLMs for generating realistic data, information theory for assessing feature importance, PCA for discovering hidden structures, and Bayesian inference for state estimation provides a complete pipeline for brain signal processing. This framework also holds the potential for generalization to more complex 2D and 3D environments.